

CJM410 Wi-Fi Module Datasheet

Ver. 1.5

Revision History

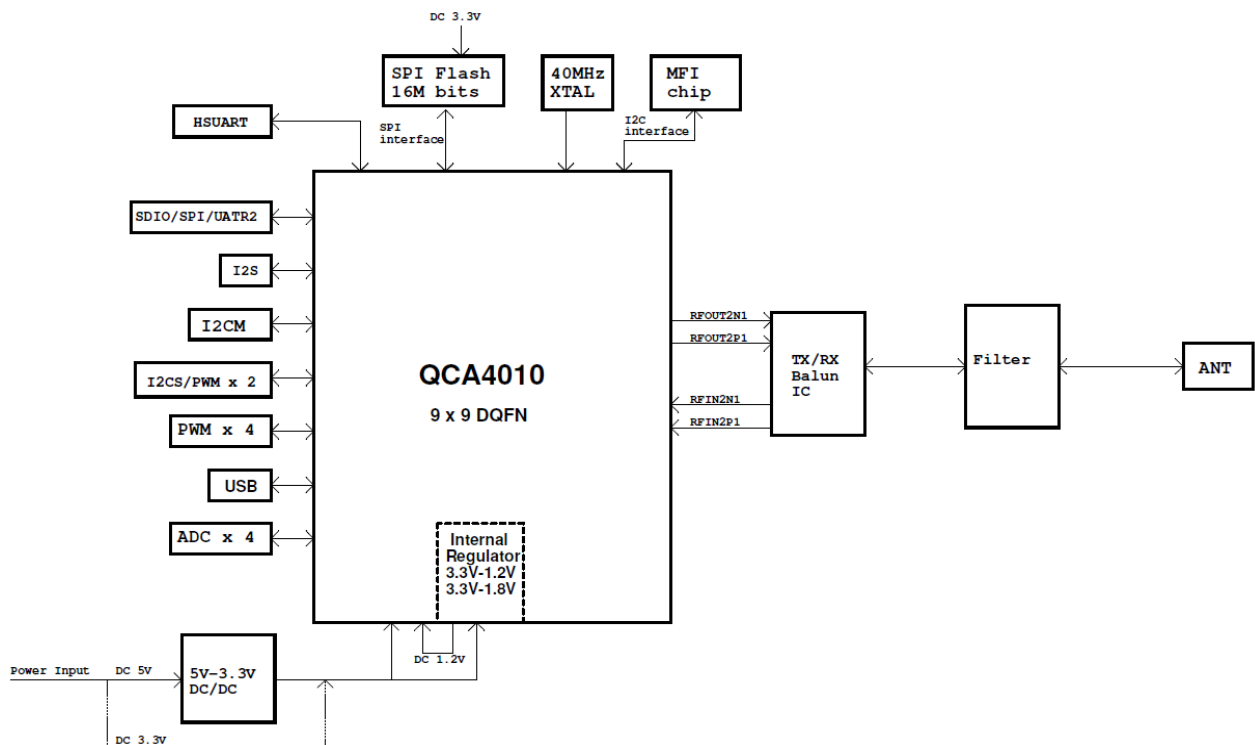
Revision	Date	Approved	Description
1.0	June 2016	Anderson	Initial release
1.1	Nov. 2016	Anderson	Add power management states and electrical characteristics
1.2	Jan. 2017	Anderson	Update operating temperature spec.
1.3	Apr. 2017	Eileen	Add radio Rx and Tx characteristics Add placement and layout information. Add manufacture information
1.4	Oct. 2019	Anderson	Update power consumption performance
1.5	November 2021	Anderson	Add SPI interface timing

1. General Description.....	4
2. Block Diagram.....	4
3. Module Pictures.....	4
4. Hardware Specifications.....	5
5. Mechanical.....	6
6. Module Pinout Definition.....	7
7. Power Management States.....	9
8. Electrical Characteristics.....	10
8.1 General DC electrical characteristics.....	10
8.2 Power-On-Reset timing.....	11
8.3 Power consumption.....	11
8.4 Rx characteristics for 2.4GHz operations.....	13
8.5 Tx characteristics for 2.4GHz operations.....	13
8.6 Timing specifications.....	14
8.6.1 SPI slave interface timing.....	15
8.6.2 SPI master interface timing.....	15
9. Module Placement and Layout Guideline.....	16
9.1 Placement on baseboard.....	16
9.2 Layout on baseboard top layer.....	17
10. Manufacture Information.....	18
10.1 Soldering recommendations.....	18
10.2 Handling information.....	18

1. General Description

The CJM410 Wi-Fi module provides quick, easy and cost effective way to enable WiFi connectivity for low-power embedded applications. The module is based on Qualcomm QCA4010 SoC which is a single band 2.4GHz 802.11 b/g/n with single-stream capability for both Tx and Rx. It has an integrated network processor with a large set of TCP/IP with IPv4/IPv6-based services. These services can be accessed via a serial SPI link or a UART link connected to an external host CPU. The CJM410 module can be either used with MP1-base board development kit for software development or incorporated into OEM products to enable rapid deployment of Wi-Fi connected systems.

2. Block Diagram



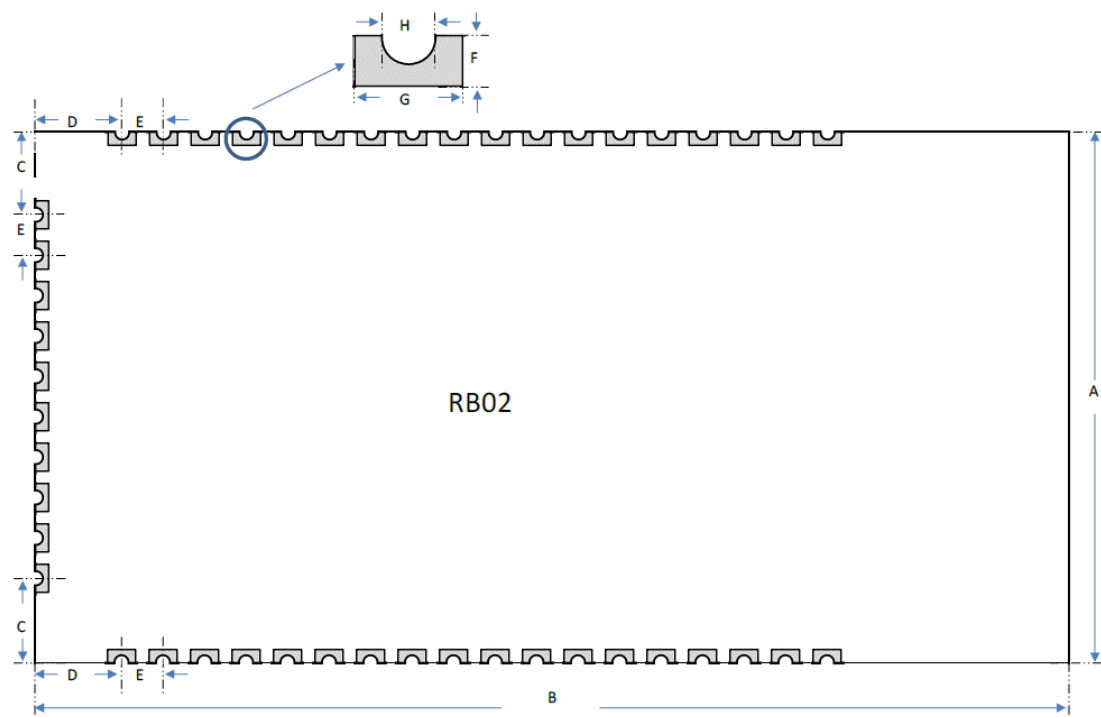
3. Module Pictures



4. Hardware Specifications

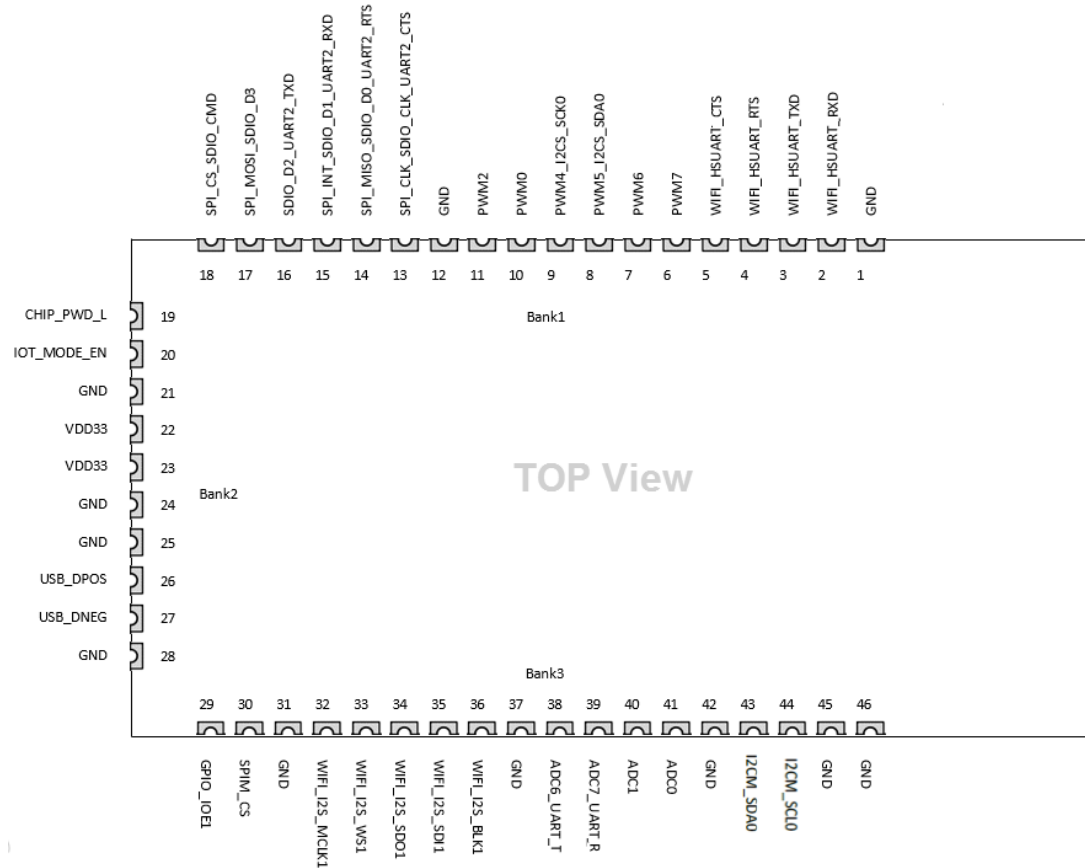
RF characteristics	
Standard	IEEE 802.11 b/g/n 1x1 Wi-Fi radio
Operating Frequency	2.412 GHz ~ 2.484 GHz
Supported Data Rates	802.11n: 6.5Mbps ~ 65Mbps (MCS 0 to 7, HT20) 802.11g: 6, 9, 12, 18, 24, 36, 48, 54 Mbps 802.11b: 1, 2, 5.5, 11 Mbps
Transmit Power (+/- 1.5dBm)	802.11b: 18dBm 802.11g: 18dBm@6Mbps, 14dBm@54Mbps 802.11n: 13dBm@MCS7
Receiver Sensitivity	802.11b: <-76dBm 802.11g: <-65dBm 802.11n: <-64dBm
Antenna Options	Printed antenna or MHF RF connector (optional)
Hardware Characteristics	
Chipset	Qualcomm Atheros QCA4010 SoC with Xtensa processor, up to 130 MHz
Memory	RAM: 1.4 MB, external SPI flash : 2 MB
I/O Interfaces	- SPI Master/Slave x 1, debug UART x 1 - High speed UART x 2: up to 3Mbps - I2C master x 1, I2C slave x 1, I2S x 1, PWM x 6, ADC x 4
Operating Temperature	0°C ~ 85°C
Storage Temperature	-40 °C ~ 135°C
Power source	3.3 V I/O supply voltage
Power Consumption	11.85 uA ~ 249.9 mA
Dimension	30(l) mm x16(w) mm x 3.6(h) mm

5. Mechanical



Label	Dimension (mm)
A	16
B	30
C	2.285
D	2.54
E	1.27
F	0.4
G	0.7
H (diameter)	0.5
Module height (including the RF shield)	2.6
Total height (with a coax cable plugged into the U.FL connector)	3.6

6. Module Pinout Definition



Pins Description

Pin	Signal/Interface	ALT1	ALT2	ALT3	GPIO No.
1	GND	Ground			
2	WIFI_HSUART_RXD	High speed UART RXD			GPIO[24]
3	WIFI_HSUART_TXD	High speed UART TXD			GPIO[23]
4	WIFI_HSUART_RTS	High speed UART RTS			GPIO[22]
5	WIFI_HSUART_CTS	High speed UART CTS			GPIO[21]
6	PWM7	PWM7			GPIO[13]

7	PWM6	PWM6			GPIO[12]
8	PWM5_I2CS_SDA0	PWM5	I2C Slave SDA0		GPIO[11]
9	PWM4_I2CS_SCK0	PWM4	I2C Slave SCK0		GPIO[10]
10	PWM0	PWM0			GPIO[6]
11	PWM2	PWM2			GPIO[8]
12	GND	Ground			
13	SPI_MISO_SDIO_D0_UART2_RTS	SPI MISO (master or slave)	SDIO Data0	UART RTS	GPIO[4]
14	SPI_CLK_SDIO_CLK_UART2_CTS	SPI CLK (master or slave)	SDIO CLK	UART CTS	GPIO[5]
15	SPI_INT_SDIO_D1_UART2_RXD	SPI Interrupt (slave)	SDIO Data1	UART RXD	GPIO[3]
16	SDIO_D2_UART2_TXD		SDIO Data2	UART TXD	GPIO[2]
17	SPI_MOSI_SDIO_D3	SPI MOSI (master or slave)	SDIO Data3		GPIO[1]
18	SPI_CS_SDIO_CMD	SPI CS (master or slave)	SDIO Command		GPIO[0]
19	CHIP_PWD_L	Module reset, active low			
20	IOT_MODE_EN	Wakeup manager enable			
21	GND	Ground			
22	VDD33	3.3V power supply			
23	VDD33	3.3V power supply			
24	GND	Ground			
25	GND	Ground			
26	USB_DPOS	USB Data+			
27	USB_DNEG	USB Data-			
28	GND	Ground			
29	GPIO_IOE1	external wakeup			
30	SPIM_CS	Flash memory /CS pin			GPIO[35]
31	GND	Ground			
32	WIFI_I2S_MCLK1	I2S MCLK1			GPIO[33]

33	WIFI_I2S_WS1	I2S WS1			GPIO[32]
34	WIFI_I2S_SDO1	I2S SDO1			GPIO[31]
35	WIFI_I2S_SDI1	I2S SDI1			GPIO[30]
36	WIFI_I2S_BLK1	I2S BLK1			GPIO[27]
37	GND	Ground			
38	ADC6_UART_T	ADC6	Debug UART TXD		GPIO[29]
39	ADC7_UART_R	ADC7	Debug UART RXD		GPIO[28]
40	ADC1	ADC1			
41	ADC0	ADC0			
42	GND	Ground			
43	I2CM_SDA0	I2C Master SDA0			GPIO[25]
44	I2CM_SCL0	I2C Master SCL0			GPIO[26]
45	GND	Ground			
46	GND	Ground			

Bootstrap signals

Pin No.	Bootstrap name	Description	
11	Test mode enable	Should be low while reset released, for normal function	
18	Host mode[1]	Bootstrap for host interface selection. Default mode is 00.	
13	Host mode[0]		
		00	USB/manufacturing test and configuration/hostless
		01	Hostless (serial AT command) mode
		10	SPI host mode
		11	SDIO host mode
20	IOT mode enable	Keep high always, for normal function	

7. Power Management States

State	Description
POWER_DOWN	CHIP_PWD_L pin assertion immediately brings the chip to this state.
	Sleep clock is disabled.
	No state is preserved.
SUSPEND	While in suspend state, the chip shuts down all circuits except a few critical

	blocks needed to resume operation after suspend.
HOST_OFF	Network services and WLAN are off. Only the SPI host interface is powered on, the rest of the chip is power gated (off).
	The host can transition QCA4010 to WAKEUP (followed by ON) at any time by writing a register in the host interface domain.
	WLAN and CPU states are not retained.
	For UART hosted, or USB manufacturing configurations, this state is bypassed by pulling GPIO0 low at the de-assertion of CHIP_PWD_L. This state applies only to SPI designs.
SLEEP	Only the sleep clock is operating.
	The crystal or oscillator is disabled.
	Any wakeup events (MAC, host, LF timer, GPIO interrupt force a transition to WAKEUP.
	All internal states are maintained.
	Host interface is idle (USB is in SUSPEND).
WAKEUP	The system transition from sleep OFF states to ON.
	The high frequency clock is gated off as the oscillator is brought up and the PLL is enabled.
	WAKEUP duration is less than 2 ms.
ON	The high speed clock is operational.
	Lower-level clock gating is implemented at the block level, including the CPU, which can be gated off using WAITI instructions while the system is on.

8. Electrical Characteristics

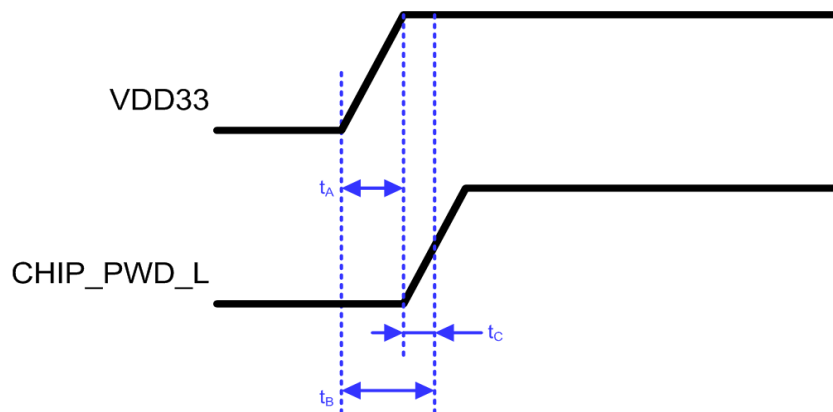
8.1. General DC electrical characteristics

These conditions apply to all DC characteristics unless otherwise specified: $T_{amb} = 25^{\circ}\text{C}$, $V_{DD33} = 3.3\text{V}$

DC electrical characteristics for digital I/Os

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
V_{IH}	High level I voltage	-	1.8	-	3.6	V
V_{IL}	Low level I voltage	-	-0.3	-	0.3	V
V_{OH}	High level O voltage	-	2.2	-	3.3	V
V_{OL}	Low level O voltage	-	0	-	0.4	V

8.2. Power-On-Reset timing



Parameter	Description	Min	Max	Unit
t_A	Rise time of VDD33 to 90% of 3.3 V	--	25	ms
t_C	Time from VDD33 reaching 90% of 3.3 V to the level of CHIP_PWD_L going above $0.5 \times VDD33$	5	--	μs
t_B	The value is $t_A + t_C$; during this time, the level of CHIP_PWD_L should stay below $0.5 \times VDD33$			

8.3. Power consumption

8.3.1. Typical power consumption for continuous Rx

Mode/Rate (Mbps)	Current consumption (mA)
Rx 11 Mbps	62
Rx HT20 MCS0	63.8
Rx HT40 MCS0	73.6

8.3.2. Typical power consumption for continuous Tx

Mode/Rate (Mbps)	Tx power (dBm)	Current consumption (mA)
Tx 1 Mbps	6.3	130.3
	8.1	159.0
	10	164.7
	12	193.5
	14	207.1

	16	223.3
	18	246.5
Tx 6 Mbps	6	132.1
	8..4	139.5
	10.4	166.5
	12.3	175.4
	14.4	204.9
	16.4	224.1
	18.2	246.8
Tx HT20 MCS0	6.4	133.6
	8.3	141.1
	10.3	168.7
	12.3	177.5
	14.3	207.7
	16.3	227.1
	18.2	249.9

8.3.3. Typical power consumption for low power states

Mode	State	Typical current consumption, hosted (11ng HT20)	Unit	Note
Standby	Suspend	11.85	μA	
	Sleep (between beacons)	201.54	μA	
DTIM1	PS enabled, LPL disabled	1.09	mA	
DTIM 3		562.77	μA	
DTIM 5		411.23	μA	
DTIM10		347.31	μA	
Connection idle	PS disabled	60.39	mA	
Active	TCP uplink	120.48 (17557 kbps)	mA	
	TCP downlink	79.9 (20352 kbps)	mA	
Active (UDP uplink)	Green Tx enabled	142.27 (41320 kbps)	mA	
	Green Tx disabled	174.6 (41640 kbps)	mA	
Active (UDP downlink)	LPL enabled	79.08 (25769 kbps)	mA	
	LPL disabled	89.34 (29238 kbps)	mA	

8.4 Rx characteristics for 2.4GHz operation

Symbol	Parameter	Conditions ¹	Min	Typ	Max	Unit
F _{Rx}	Rx input frequency range	–	2.412	–	2.472	GHz
S _{rf}	Sensitivity					
	CCK	1 Mbps	–	-93	–	dBm
		11 Mbps	–	-87	–	
	OFDM	6 Mbps	–	-89	–	
		54 Mbps	–	-73	–	
	HT20	MCS0	–	-89	–	
		MCS7	–	-70	–	
R _{adj}	Adjacent channel rejection					
	CCK	2 Mbps	–	47	–	dB
	OFDM	6 Mbps	–	36	–	
		54 Mbps	–	21	–	
	HT20	MCS0	–	34	–	
		MCS7	–	18	–	

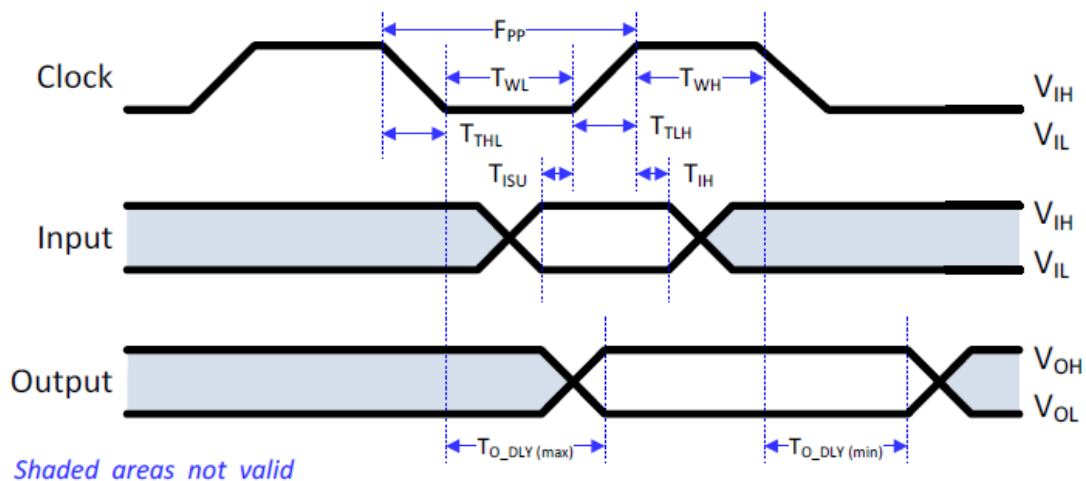
1. In LPL mode, sensitivity will be degraded by 1 – 2 dB.

8.5 Tx characteristics for 2.4GHz operation

Symbol	Parameter	Conditions	Min	Typ	Max	Unit
F _{tx}	Tx output frequency range	–	2.412	–	2.472	GHz
P _{out}	Output power¹					
	802.11b mask compliant	1 Mbps	–	19	–	dBm
	802.11g mask compliant	6 Mbps	–	19	–	
	802.11g EVM compliant	54 Mbps	–	16	–	
	802.11n HT20 mask compliant	MCS0	–	19	–	
	802.11n HT20 EVM compliant	MCS7	–	15	–	
1. Refer to IEEE802.11 specification for Tx spectrum limits: ▫ 802.11b mask (18.4.7.3) ▫ 802.11g mask (19.5.4) ▫ 802.11g EVM (17.3.9.6.3) ▫ 802.11n HT20 mask (20.3.21.1) ▫ 802.11n HT20 EVM (20.3.21.7.3)						

8.6 Timing specifications

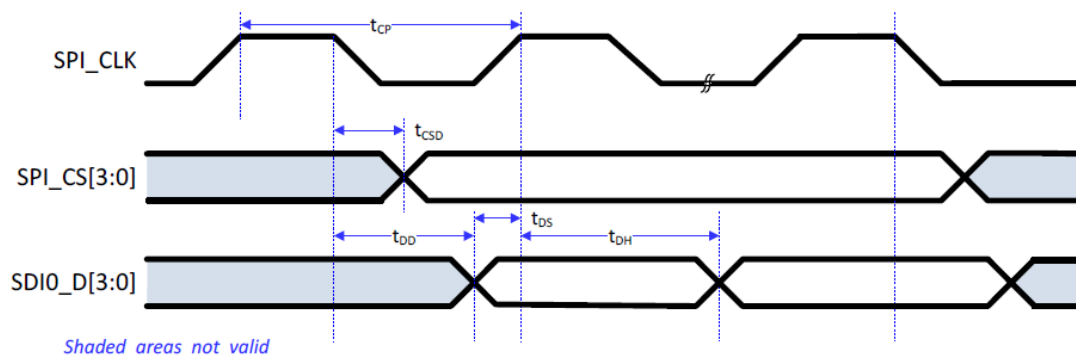
8.6.1 SPI slave interface timing



SPI slave timing constraints (48MHz reference clock)

Parameter	Description	Min	Max	Unit
f_{PP}	Clock frequency	0	48	MHz
t_{WL}	Clock low time	8.3	-	ns
t_{WH}	Clock high time	8.3	-	ns
t_{TLH}	Clock rise time	-	2	ns
t_{THL}	Clock fall time	-	2	ns
t_{ISU}	Input setup time	5	-	ns
t_{IH}	Input hold time	5	-	ns
T_{O_DLY}	Output delay	0	5	ns

8.6.2 SPI master interface timing



SPI master timing constraints

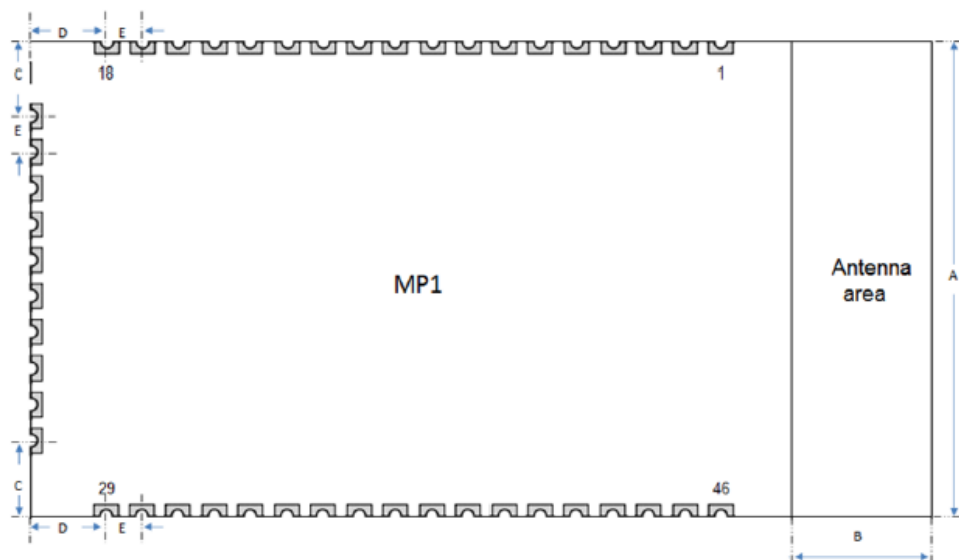
Parameter	Description	Min	Max	Unit
t_{CP}	Clock period	30.7	1000	ns
t_{CSD}	Chip select valid delay	-5.5	5	ns

t_{DD}	Data valid delay	-5.5	5	ns
t_{DS}	Data setup	3	-	ns
t_{DH}	Data hold	0	-	ns

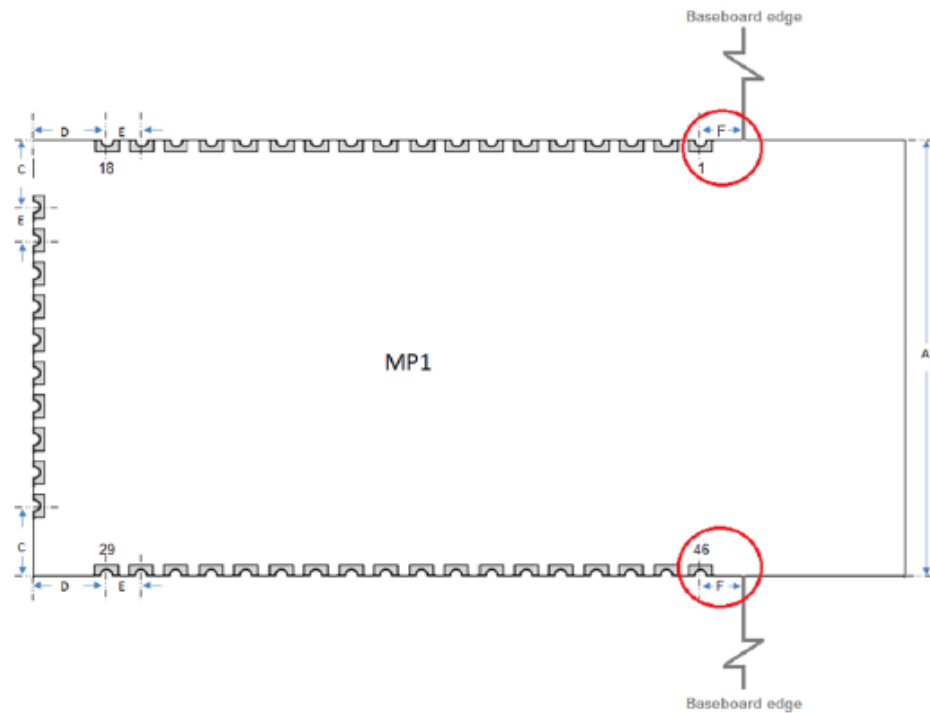
9. Module Placement and Layout Guideline

9.1 Placement on baseboard

- The Antenna area should be outside of the baseboard.
- No metal material cover the Antenna area after module fixed on final production.
- The Antenna area : A=16 mm, B=3.8 mm

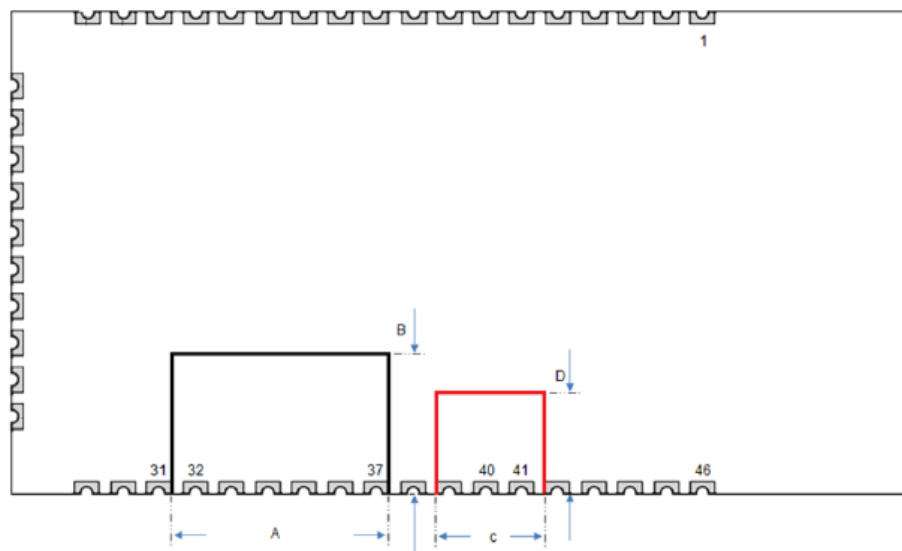


- Distance from Pin 1 & Pin 46 of module to the edge of baseboard should be less than 1.7 mm



9.2 Layout on baseboard top layer

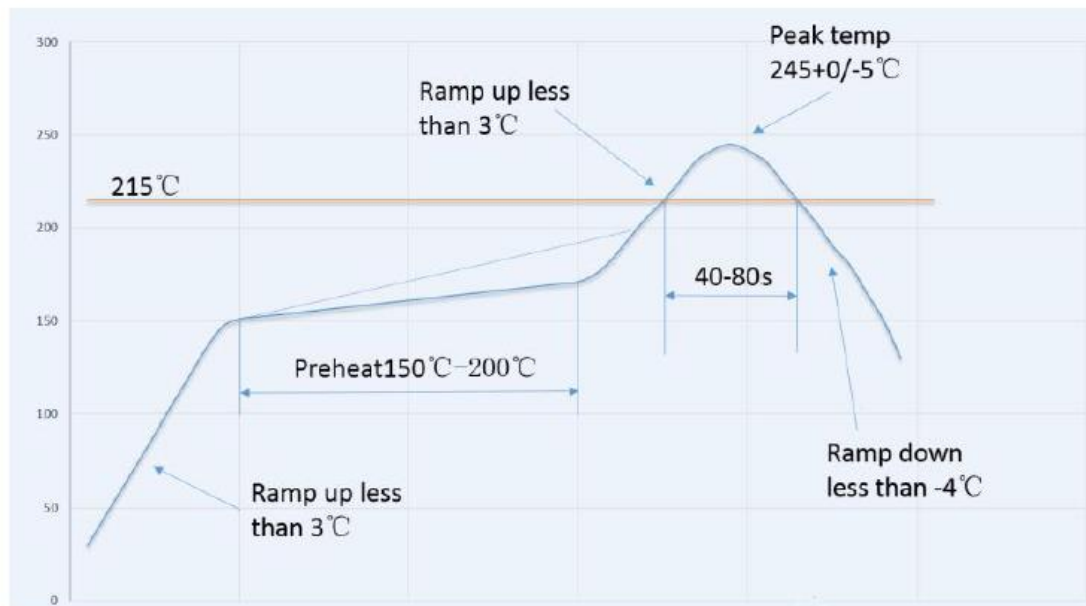
- Do not route any trace on top layer of baseboard in the black & red rectangle to avoid crosstalk
 - There is I2S signal in the black rectangle area on module bottom layer.
A=7.7mm, B=4.8mm
 - There is ADC signal in the red rectangle area on module bottom layer
C=3.9mm, D=3.3mm



10.Manufacture Information

10.1 Soldering recommendations

The module can be SMT on the board following the below temperature curve graph.



10.2 Handling information

The module contains a highly sensitive electronic circuitry. Handling without proper ESD protection will destroy or damage the chip permanently.